JUNO: NEW ORIGINS / SIMPLEROCKETS 2

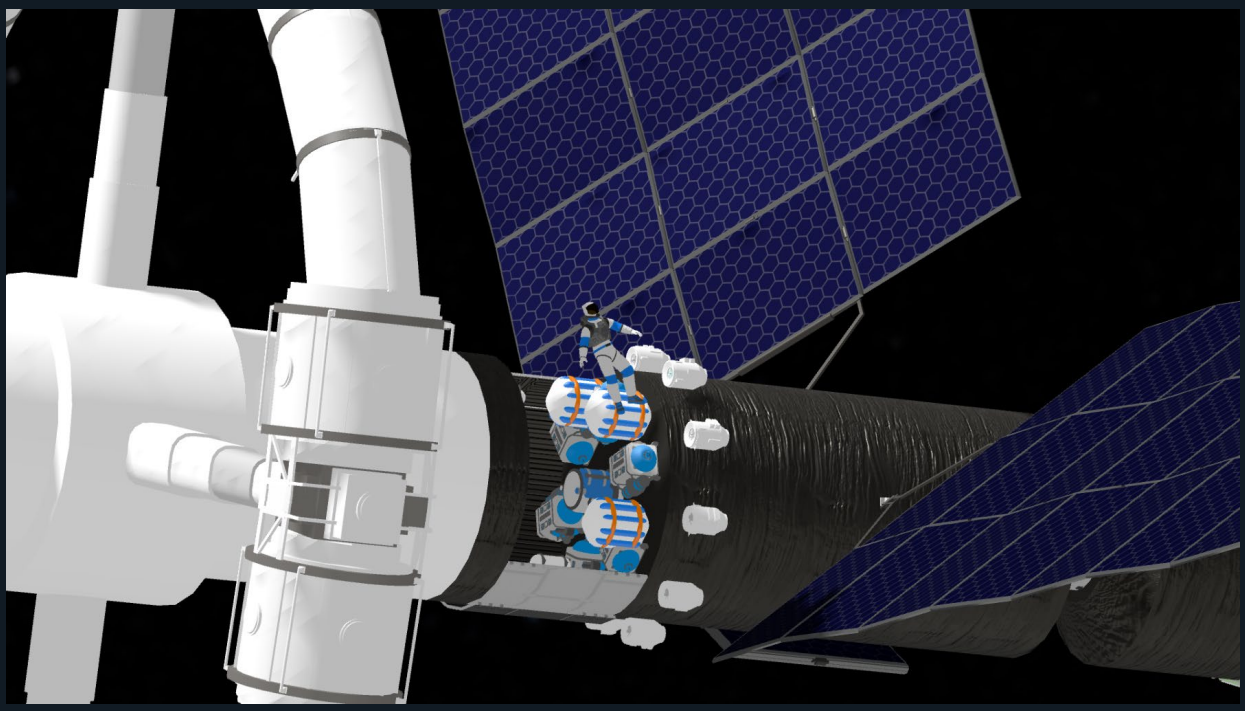
DROODISM

用户手册
USER MANUAL

中英双语 / CHINESE + ENGLISH

让每一次载人任务都有补给、有保障、有归途。

Plan supplies. Protect the crew. Bring them home.



图像源自手册 3.0 / Image from manual 3.0

手册 / MANUAL 4.0　　模组 / MOD 0.89

SatelliteTorifune

2026-10-24

生命维持 · 资源循环 · 乘员管理 · 辐射防护

LIFE SUPPORT / RESOURCE CYCLES / CREW / RADIATION

阅读指南与目录
Reader's guide and contents

左侧中文，右侧英文；表格按双语合并。章节号可点击，PDF 书签可快速跳转。

Chinese is on the left, English on the right. Tables combine both languages. Click a chapter or use PDF bookmarks.

章节 / CHAPTER			页 / PAGE
01	安装与开始	Install and get started	3
02	六种维生资源	The six life support resources	4
03	按人数和时间规划	Size supplies for the mission	5
04	乘员、出舱与呼吸	Droods, EVA and breathing	6
05	乘员职业与插旗	Crew roles and flag planting	7
06	治疗与零件维修	Healing and part repairs	8
07	个人降落伞	Personal parachutes	9
08	储箱与资源转移	Storage and resource transfer	10
09	制氧与二氧化碳处理	Oxygen production and CO2 removal	11
10	水、污水与取水	Water, wastewater and collection	12
11	资源包与应急备份	Resource packs and emergency backup	13
12	光生物反应器	The photobioreactor	14
13	气体加压与减压	Gas compression and decompression	15
14	燃料电池的维生副产品	Life support from fuel cells	16
15	冬眠舱	Hibernating chambers	17
16	原位资源利用与燃料合成	ISRU and fuel synthesis	18
17	重力环与气球	Gravity rings and balloons	19
18	阅读飞行检查面板	Read the flight inspector	20
19	资源预警与时间加速	Resource alerts and time warp	21
20	地图中的辐射带	Radiation belts in map view	22
21	舱体辐射屏蔽	Compartment radiation shielding	23
22	剂量、伤害与船内辐射	Dose, damage and onboard sources	24
23	供电与 RTG 衰减	Power and RTG decay	25
24	模组设置速查	Mod settings reference	26
25	配置与自定义旗帜	Configuration and custom flags	27
26	读档、切船与后台消耗	Saves, switching and background use	28
27	常见问题排查	Troubleshooting	29
28	任务前后的检查清单	Mission checklists	30
29	附录：有机分类与回收装置	Appendix: the DOCR device	31
30	零件与章节索引	Parts and chapter index	32
31	版本说明与致谢	Edition notes and credits	33

01 安装与开始
Install and get started

中文

1. 安装依赖

先安装并启用 Juno Harmony，再启用 Droodism。打开游戏的模组管理器，确认两个模组均已启用，且 Droodism 没有初始化错误。

2. 先做一次地面测试

建议使用新存档开始。为一名乘员加上资源包，在发射台打开生命维持面板，确认氧气、水和食物可用，废物有储存空间。先以正常速度观察，再使用时间加速。

3. 查看更新提示

模组会在每次游戏会话中检查一次更新。发现更高版本后，可在主菜单选择下载更新、稍后再说或不再提醒。不再提醒仅跳过该版本。

ENGLISH

1. Install the dependency

Install and enable Juno Harmony before enabling Droodism. In the game's mod manager, confirm that both mods are enabled and Droodism reports no initialization error.

2. Run a ground test

A new save is recommended. Add a resource pack for one crew member, launch, and open the life support inspector. Confirm that oxygen, water and food are available and waste has storage space. Observe at normal speed before using time warp.

3. Check update notifications

The mod checks for updates once per game session. When a newer version is found, the main-menu dialog offers Download Update, Later and Dismiss. Dismiss skips that version; a later version may notify you again.

入口 / Destination	链接 / Link
Juno Harmony	Juno Harmony mod page
项目与发布 / Project and releases	SatelliteTorifune / Droodism releases
示例飞船 / Example craft	Life Support Mod Demo Craft

> 已有存档请先备份。不要在唯一的任务存档上直接试验模组升级或卸载。
>
> Back up existing saves before upgrading or removing the mod, and test with a copy of an ongoing mission.

02 六种维生资源
The six life support resources

中文

补给会减少，废物会增加

每名宇航员和游客都需要氧气、食物和水，并产生二氧化碳、废水和固体废物。缺少补给或废物储满都会造成伤害，最终可能导致乘员死亡。

ENGLISH

Supplies fall; waste rises

Astronauts and tourists need oxygen, food and water, and produce carbon dioxide, wastewater and solid waste. Empty supplies or full waste storage cause damage and can eventually kill the crew.

资源 / Resource	用途与处理 / Purpose and handling
气态氧气 / Oxygen	供乘员呼吸；由电解、减压等设备补充。 Breathing supply; replenish by electrolysis, gas decompression or other producers.
食物 / Food	随船携带，或由光生物反应器分批生成。 Carry reserves or grow batches in a photobioreactor.
水 / H2O	供乘员及设备使用；可回收废水或在水中取水。 Used by crew and equipment; recover wastewater or collect water when submerged.
气态二氧化碳 / LPCO2	储存、过滤排出，或作为其他设备的原料。 Store, vent through a filter, or feed another device.
废水 / Wasted Water	需要空余储存空间；可送入污水处理装置。 Needs storage headroom; feed it to a sewage treatment device.
固体废物 / Solid Waste	需要空余储存空间；可促进光生物反应器生长。 Needs storage headroom; can boost photobioreactor growth.

注意气体类型

气态氧气 Oxygen 与液氧 LOX 是不同资源；气态二氧化碳 LPCO2 与高压二氧化碳 CO2 也不同。装满液氧并不等于乘员已有可呼吸的氧气。

Match the gas type

Breathable Oxygen and liquid oxygen (LOX) are different resources. Low-pressure LPCO2 and high-pressure CO2 are also separate. A full LOX tank alone does not provide breathable oxygen.

> 资源不足、废物与辐射造成的伤害受游戏的撞击伤害倍率影响；该倍率为零时，对应伤害也被关闭。
>
> Supply, waste and radiation damage use the game's impact-damage multiplier. Setting that multiplier to zero also disables these damage paths.

03 按人数和时间规划
Size supplies for the mission

中文

先计算乘员天数

把每人的任务时长相加。三名普通乘员飞行两天，相当于六个乘员天。资源包按单名普通乘员约 24 小时的补给配置设计；还应为等待、对接、返航和救援留出余量。

ENGLISH

Start with crew-days

Add up the mission duration for each crew member. Three ordinary crew flying for two days need six crew-days. The resource pack is designed around roughly 24 hours for one ordinary crew member; add margin for delays, docking, return and rescue.

默认基准 / Default baseline	每人每 24 小时 / Per crew per 24 h
氧气消耗 / Oxygen used	604.8 L
水消耗 / Water used	约 / about 3.00 L
食物消耗 / Food used	约 / about 0.501 L
气态 CO2 生成 / LPCO2 produced	约 / about 249.5 L
废水生成 / Wastewater produced	约 / about 0.989 L
固体废物生成 / Solid waste produced	约 / about 0.00661 L

这些是规划基准

表中使用游戏资源体积单位，按默认倍率、非冬眠、使用内部氧气的普通乘员计算。游客消耗乘以 1.05；被判定为跑动时再乘以 1.75。可呼吸环境会改变氧气与二氧化碳的收支。

Use the baseline with care

Values use in-game resource volume, default rates, internal oxygen and a non-hibernating ordinary crew member. Tourists multiply consumption by 1.05; running multiplies it by another 1.75. Breathable environments change the oxygen and CO2 balance.

> 循环设备不能替代出发时的储备。食物分批生长、夜间失电和废物储满都可能打断循环。
>
> Carry departure reserves even with recycling equipment. Batch food growth, loss of power at night and full waste storage can interrupt the loop.

04 乘员、出舱与呼吸
Droods, EVA and breathing

中文

ENGLISH



设计器中的乘员与属性面板 / Drood and designer properties
旧版示意，源自手册 3.0 / Historical illustration, manual 3.0

出发前设置随身补给

选择乘员，在生命维持属性中调整氧气、食物和水的携带量。增加携带量会增加质量，但不会同时扩大随身废物容量。默认补给量约为普通乘员静止消耗的 12 小时，实际受环境和活动影响。

Set carried supplies before launch

Select a Drood and adjust oxygen, food and water in the life support properties. More supplies add mass without enlarging personal waste capacity. Default carried supplies cover roughly 12 hours at ordinary resting rates, depending on conditions and activity.

登船与出舱

在舱内可利用所在飞船的共享资源。登船或资源引用刷新时，模组会尝试补满随身补给并把废物转入飞船；飞船必须有补给和空余废物容量。出舱后要同时查看三种补给与三种废物。

Boarding and EVA

While aboard, a Drood can use the craft's shared resources. Boarding or refreshing resource links attempts to refill personal supplies and offload waste; the craft needs both supplies and free waste capacity. During EVA, monitor all three supplies and all three wastes.

何时可以呼吸外界空气

大气密度必须高于 0.4，且行星在名单中，或大气组分满足氧气判定。默认名单为 Droo、Earth、Kerbin、Laythe；组分判定要求 LOX 质量分数至少 10%。使用外界氧气时会补充随身氧气，并跳过正常 CO2 生成；水下或高空稀薄空气仍需内部氧气。

When outside air is breathable

Air density must exceed 0.4, with a listed planet or a qualifying atmosphere. The default list is Droo, Earth, Kerbin and Laythe; composition requires at least 10% LOX mass fraction. Outside-air use refills carried oxygen and skips normal CO2 production. Underwater or thin high-altitude air still requires internal oxygen.

05 乘员职业与插旗
Crew roles and flag planting

中文 ENGLISH

职业 Role	当前能力 / Current ability
飞行员 / Pilot	过载伤害减轻 40%；可在地表插旗。 40% reduction in G-force damage; can plant flags.
工程师 / Engineer	使用有限的修复工具修理附近受损零件。 Repairs nearby damaged parts using a finite tool supply.
科学家 / Scientist	当前没有单独的职业技能。 No separate role-specific ability is implemented in this version.
游客 / Tourist	维生消耗为普通乘员的 1.05 倍。 Uses life support supplies at 1.05 times the ordinary rate.

职业从乘员记录读取

在设计器分配乘员后，核对其姓名和职业；飞行中也可查看。职业、累计辐射与任务时间关联到存档中的乘员记录。未分配乘员或没有对应记录时，属性可能显示“未知”，不能据此假定已有职业技能。

Roles follow crew records

After assigning crew in the designer, check their names and roles; they are also shown in flight. Roles, cumulative radiation and mission time are tied to saved crew records. Unassigned crew or missing records may show Unknown; do not assume role abilities are available in that state.

让飞行员插旗

让飞行员出舱并站稳在地面，速度低于 1 m/s，且不在水中。打开其检查面板并点击“插旗”。当前资源限制通常只允许该飞行员执行一次插旗。旗面可使用自定义图片，路径见第 25 章。

Plant a flag with a pilot

Put a pilot on EVA, grounded, moving below 1 m/s and out of water. Open the pilot's inspector and choose Plant Flag. The current allowance normally permits one flag per pilot. A custom flag image can be supplied; see chapter 25.

飞行员的加成只针对过载伤害，不会消除缺氧、废物或辐射造成的伤害。

The pilot bonus applies to G-force damage. It does not remove damage from starvation, waste buildup or radiation.

06 治疗与零件维修 Healing and part repairs

中文

用急救包治疗乘员

保持急救包的检查面板可用，选择附近受伤的 Drood，然后点击“治疗 Drood”。目标与急救包必须保持在 5 m 内，治疗会消耗急救包的治疗点数。目标改变、距离过远或点数耗尽都会使治疗中断。

先解除持续伤害条件

乘员随身的氧气、食物和水都不能为空，三种废物都不能储满；累计辐射达到 800 rad 后也无法接受治疗。急救包恢复健康，不会清除累计辐射。

工程师修理零件

打开工程师的检查面板，选择 5 m 内受损的非乘员零件，再点击“修理”。修复过程会消耗工程师携带的工具。修理功能不能用来治疗另一名 Drood。

用工具箱补给

保持工具箱面板可用并选择 5 m 内的工程师，点击“补充工程师工具包”。工具箱本身的点数会减少。任务前同时检查工程师工具量与工具箱库存。

急救包用于治疗乘员，工程师用于修理零件；两者的目标、点数与操作按钮不同。

ENGLISH

Heal a Drood with a first aid kit

Keep the first aid kit inspector available, select the injured Drood, and choose Heal Drood. Stay within 5 m of the kit. Healing spends its finite healing points and stops if the target changes, moves too far away or the kit runs out.

Remove the cause of damage first

Personal oxygen, food and water must be non-empty, and none of the three waste buffers may be full. At 800 rad cumulative dose, healing is also blocked. First aid restores health; it does not erase cumulative radiation.

Repair parts with an engineer

Open an engineer's inspector, select a damaged non-Drood part within 5 m, and choose Repair. Repairs consume the engineer's tools. This action cannot heal another Drood.

Resupply from a tool kit

Keep the tool kit inspector available, select an engineer within 5 m, and choose Refill Tool Kits. This consumes the tool kit's own supply. Check both the engineer's tools and the kit's reserve before launch.

First aid kits heal crew, while engineers repair parts. They use different targets, supplies and action buttons.

07 个人降落伞 Personal parachutes

中文

在设计器选择伞型

乘员属性提供不携带、普通降落伞和滑翔伞选项，并可设置自动开伞及完全展开高度。默认自动开伞关闭，自动开伞高度 500 m，完全展开高度 250 m。完全展开高度不能高于自动开伞高度。

ENGLISH

Choose the canopy in the designer

Drood properties offer None, Parachute and ParaGlider, with automatic deployment and full-deployment heights. Defaults are automatic deployment off, deployment at 500 m and full deployment at 250 m. Full-deployment height cannot exceed deployment height.

开伞条件 / Requirement	当前限制 / Current limit
状态 / State	处于舱外、未着地、未在水中。 On EVA, airborne and out of water.
离地高度 / Height	必须高于 10 m。 Must be above 10 m over terrain.
空气 / Air	空气密度必须高于 0.01。 Air density must exceed 0.01.
速度 / Speed	低于当地音速。 Below the local speed of sound.
自动触发 / Automatic trigger	正在下降，且已到设置高度以下。 Descending and below the selected deployment height.

使用前先测试

在乘员面板中使用相应的开伞控制。开伞时乘员进入伞具乘坐状态，落地后自动离伞并清理伞具；自动开伞也会被关闭。当前未提供收伞打包与重复使用流程，任务前应在目标大气条件下测试。

Test before depending on it

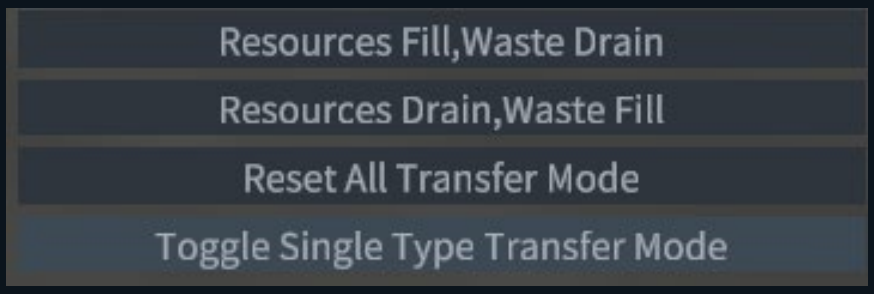
Use the deployment control in the Drood inspector. Deployment seats the Drood in the canopy; landing unloads the Drood, removes the canopy and switches off automatic deployment. This version offers no repacking-and-reuse workflow. Test in the intended atmosphere before a mission.

真空、超音速或高度过低时无法开伞。为任务设计独立的减速和着陆余量。

Deployment is blocked in vacuum, at supersonic speed or too close to the ground. Allow enough height and an appropriate descent profile.

08 储箱与资源转移
Storage and resource transfer

中文 ENGLISH



四种批量转移操作 / The four transfer shortcuts
旧版示意，源自手册 3.0 / Historical illustration, manual 3.0

同一飞船共享资源

六种维生资源按飞船共享资源处理，不需要为每台设备单独连接维生管线。分离后不再属于同一飞船的储箱不会继续共享；要检查乘员实际所在的一侧。

Resources are shared within a craft

The six life support resources are pooled at craft level, without a dedicated life support pipe to every device. Tanks on a separated craft no longer share that pool. Check the side that actually contains the crew.

面板操作 / Inspector action	结果 / Result
资源填充，废物排出 Resources Fill, Waste Drain	三种补给设为 Fill，三种废物设为 Drain。 Set the three supplies to Fill and the three wastes to Drain.
资源排出，废物填充 Resources Drain, Waste Fill	执行相反的设置。 Apply the reverse settings.
重置所有传输模式 Reset All Transfer Mode	六种资源全部设为 None。 Set all six transfer modes to None.
切换单种资源传输模式 Toggle Single Type Transfer Mode	显示各资源的 Fill / None / Drain 控制。 Reveal Fill / None / Drain controls for each resource.

方向与连接都要正确

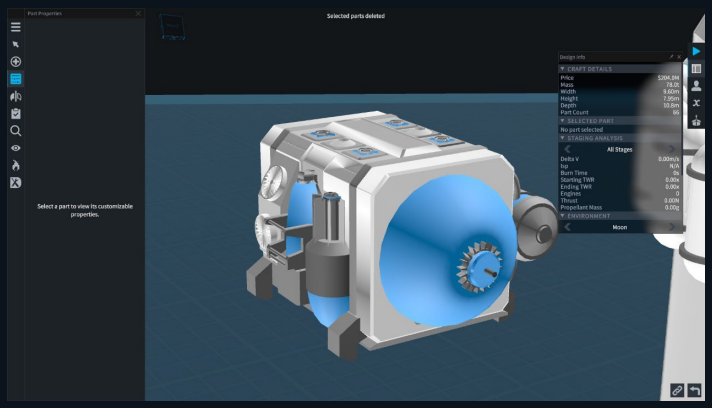
这些按钮设置转移模式，实际转移仍需要游戏支持的飞船连接与有效储箱。补给双方要设置相容方向。单独一名出舱 Drood 无法使用这些批量转移模式。

Set both direction and connection

These buttons set transfer modes; actual transfer still requires a valid connection and compatible tanks in the game. Configure compatible directions at both ends. A single EVA Drood cannot use these craft transfer shortcuts.

09 制氧与二氧化碳处理
Oxygen production and CO2 removal

中文 ENGLISH



电解水装置 / Electrolytic device
旧版示意，源自手册 3.0 / Historical illustration, manual 3.0

电解水装置

消耗水和电力，向气态氧气储箱补氧，并在有液氢储存空间时产生液氢。启动前准备水、可用电量和氧气空余容量。液氢储箱用于保存副产品；没有液氢接收空间并不阻止氧气生产。

Electrolytic Device

Consumes water and electricity, supplies breathable oxygen, and stores liquid hydrogen when an LH2 tank has room. Provide water, battery charge and free oxygen capacity. An LH2 tank captures the by-product; lack of LH2 storage does not prevent oxygen production.

二氧化碳过滤器

将飞船的气态二氧化碳排出，并按设定速率消耗电力。它用于降低废物负担，不会制造氧气，也不会把二氧化碳变成高压原料。

Carbon Dioxide Filter

Vents the craft's low-pressure carbon dioxide and draws electrical power at its configured rate. It reduces waste buildup; it does not produce oxygen or turn CO2 into high-pressure feedstock.

如何组合

短任务可携带氧气并直接过滤 CO2；长任务可用电解供氧，并选择过滤 CO2 或保留给光生物反应器、压缩设备。没有实际用途时，不要让 CO2 占满储箱。

Combine them in a system

For a short mission, carry oxygen and filter CO2. For a longer mission, use electrolysis and either vent CO2 or reserve it for a photobioreactor or compressor. Do not let unused CO2 fill the waste tanks.

10 水、污水与取水
Water, wastewater and collection

中文

ENGLISH



污水处理装置 / Sewage treatment device
旧版示意，源自手册 3.0 / Historical illustration, manual 3.0

污水处理装置

开启后消耗废水和电力，向水箱加入清水。输入废水不足、电池为空或清水箱已满时会停止处理。可在设计器中调整设备规模与转换效率。

Sewage Treat Device

When activated, consumes wastewater and electricity and adds clean water. Processing stops if wastewater is empty, the battery is empty or clean-water storage is full. Size and conversion efficiency can be adjusted in the designer.

水泵

让水泵零件实际浸入水中并保持激活，即可向当前飞船补水，同时消耗电力。它不能从干燥地面直接采水，也不是已完成的地表钻井系统。

Water Pump

Submerge the pump part and activate it to add water to the current craft while drawing electricity. It does not extract water from dry ground, and it is not a completed ground-drilling system.

燃料电池也可水下取水

使用氢氧或甲烷液氧燃料的相应原版燃料电池，获得了水下取水功能；同样要求零件入水并激活。起飞离水后不要继续把这条水源计入预算。

Fuel cells can also collect water

The supported stock fuel cells using hydrolox or methalox also gain underwater water collection when submerged and active. Once the part leaves the water, do not count that source in the mission budget.

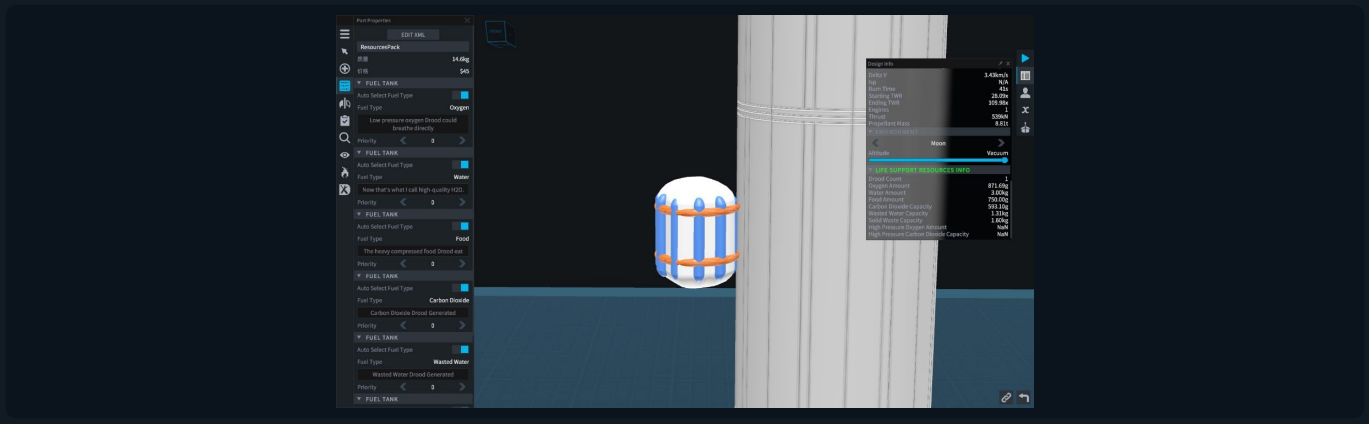
废水回收不是无限水源。为乘员、制氧、反应器和屏蔽补充共同预留清水。

Wastewater recycling is not an unlimited water source. Budget clean water for crew, oxygen production, reactors and shield refills together.

11 资源包与应急备份
Resource packs and emergency backup

中文

ENGLISH



资源包安装示例 / Example resource pack installation
旧版示意，源自手册 3.0 / Historical illustration, manual 3.0

资源包

把三种补给与三种废物储存集成在一个零件中，适合试飞和短任务。标准定义包含 610 L 氧气、3 L 水、0.5 L 食物，并预留废物空间。缩放后容量会变化，发射前以面板显示为准。

ResourcesPack

Combines the three supplies and three waste stores in one part for test flights and short missions. The standard definition contains 610 L oxygen, 3 L water and 0.5 L food, plus waste space. Scaling changes capacity; use the inspector values before launch.

过氧化钠容器

不消耗电力，通过处理二氧化碳和废水释放氧气。先激活并展开容器，反应速率受当前二氧化碳水平等参数影响。总反应量有限，用尽后不能作为持续供氧设备。

Sodium Peroxide Container

Releases oxygen while processing CO2 and wastewater without drawing electricity. Activate and deploy it first. Reaction rate depends on factors including the current CO2 level. Its reaction budget is finite, so it cannot serve as a permanent oxygen source.

备份要能独立工作

如果主制氧依赖电力，应保留直接可用的气态氧气和有限的无电备份。过氧化钠仍需要反应物和氧气接收空间；只携带一个容器不能保证任意故障下都能救援。

Make the backup independent

If primary oxygen production needs power, keep ready-to-breathe oxygen and a finite power-independent reserve. Sodium peroxide still needs reactants and oxygen storage. One container is not a guarantee against every failure.

12 光生物反应器 The photobioreactor

中文

输入与输出

消耗气态 CO2 和水，产生气态氧气及废水，并积累食物生长进度。固体废物可作为促进剂。必须有电池电量和废水空余容量；食物按进度完成后分批加入储箱。

太阳光模式

让受光面朝向太阳，检查指向效率。设备不会像自动跟踪太阳能板那样自行转向。无有效光照、缺少输入或废水储满时，生长会停滞并可能退化。即使使用太阳光，电池为空仍会阻止工作。

人工光照模式

在检查面板启用人工光照，可摆脱太阳指向要求，但会持续消耗电力。太阳光模式与人工光模式都需要事先启动、展开并保留食物库存，不能在断粮后等待第一批收成。

ENGLISH



展开后的光生物反应器 / Deployed photobioreactor
旧版示意，源自手册 3.0 / Historical illustration, manual 3.0

Inputs and outputs

Consumes low-pressure CO2 and water, produces oxygen and wastewater, and builds food-growth progress. Solid waste can boost growth. Battery charge and free wastewater capacity are required. Food is added in batches when growth completes.

Solar-light mode

Point the receiving face toward the Sun and check pointing efficiency. The device does not automatically track the Sun. Without useful light, inputs or wastewater capacity, growth stops and can decay. An empty battery also blocks operation in solar mode.

Artificial-light mode

Enable artificial light in the inspector to remove the solar-pointing requirement, at the cost of electrical power. In either mode, activate and deploy the reactor early and keep food reserves while waiting for the first batch.

13 气体加压与减压
Gas compression and decompression

中文 ENGLISH

选择 Selection	低压侧 / Low-pressure side	高密度侧 / Dense side
O2	Oxygen / 气态氧气	LOX / 液氧
CO2	LPCO2 / 气态二氧化碳	CO2 / 高压二氧化碳
N2	LPN2 / 气态氮气	N2 / 高压氮气

减压装置

把所选高密度资源转换为低压气体，正常减压不消耗电力。低压侧必须有空余容量。长任务携带液氧时，需要这条转换路径供乘员呼吸。

Gas Depressurize Device

Converts the selected dense resource into low-pressure gas without electrical consumption in normal operation. Leave free capacity on the low-pressure side. Long missions carrying LOX need this conversion path for crew breathing.

加压装置

消耗电力，把低压气体转换到高密度储箱。适合压缩要保留的 CO2，供甲烷反应器使用。两种设备都必须选择正确气体，并配备两侧储箱；转换存在损失。

Gas Pressurize Device

Uses electricity to move low-pressure gas into dense storage. It can preserve CO2 for a methalox reactor. Both devices need the correct gas selection and storage on both sides. Conversion is lossy.

紧急减压会持续排放所选类型的高、低压储气，直到排空后阀门失效；当前流程不可取消。它不是常规减压开关，也不是仅排掉少量废气。

Emergency Depressurization keeps venting both sides of the selected gas pair until empty, then disables the valve. The current sequence cannot be cancelled. This is not the normal decompression control or a small waste purge.

14 燃料电池的维生副产品
Life support from fuel cells

中文

原版燃料电池获得扩展

模组为支持的原版燃料电池添加维生副产品。需要燃料电池激活、仍有燃料并具备对应输出储箱；不要把这些产物当作任意发电方式都能获得的奖励。

ENGLISH

Stock fuel cells gain new outputs

Supported stock fuel cells gain life support by-products. The cell must be active, have fuel, and have appropriate output storage. These outputs do not apply to every type of power generator.

燃料 / Fuel	维生输出 / Life support output
氢氧 / Hydrolox (LOX/LH2)	水 + 气态氧气。 Water + breathable oxygen.
甲烷液氧 / Methalox (LOX/CH4)	水 + 气态二氧化碳。 Water + low-pressure carbon dioxide.
煤油液氧 / Kerolox (LOX/RP1)	气态二氧化碳。 Low-pressure carbon dioxide.
航空燃料 / Jet	气态二氧化碳。 Low-pressure carbon dioxide.

保留或排出 CO2

使用会产生 CO2 的燃料时，可在燃料电池检查面板中切换“将 CO2 排放到外部”。排放开启时不把这部分 CO2 收集进废物储箱；若要供生物或化学系统使用，则保留并配置相应处理链。

Store or vent CO2

For CO2-producing fuels, use Release CO2 Externally in the cell's inspector. With venting enabled, that CO2 is not collected in waste storage. To reuse it in biological or chemical systems, retain it and provide the necessary processing chain.

预算同时受到燃料限制

用燃料电池同时保障电力和制水时，一次燃料耗尽会失去多条供应。保留独立的电池、清水和可呼吸氧气，覆盖系统切换与返航时间。

The fuel supply is also a limit

If fuel cells provide both electricity and water, one depleted fuel supply can remove several services at once. Keep independent battery, water and breathable oxygen reserves for changeover and return.

15 冬眠舱 Hibernating chambers

中文

有人且有电时进入冬眠

把乘员放入冬眠舱并保持电池有电，舱内乘员会进入冬眠，不再进行正常补给消耗和废物生成。冬眠舱持续消耗大量电力；电池耗尽后，正常维生消耗恢复。

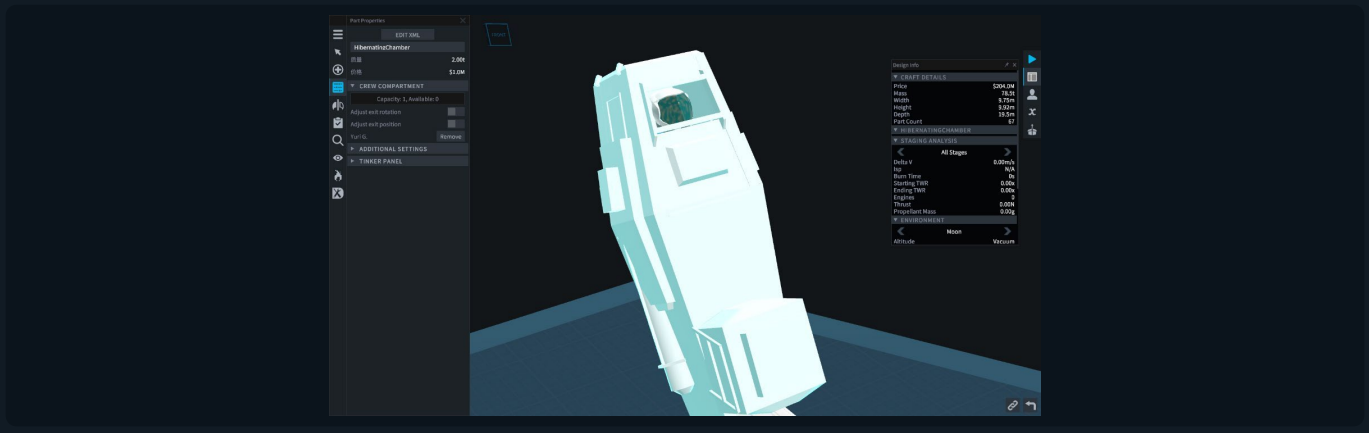
冬眠不会停止辐射

当前版本仍会计算冬眠乘员的辐射剂量与辐射伤害。进入冬眠不等于获得辐射免疫；仍需检查舱体屏蔽、任务路线与累计剂量。

为失电准备退路

冬眠方案应保留醒来后的应急补给，确认最长无光期的电量，并为长期任务计入 RTG 功率衰减。跨飞船切换或读档后，重新检查冬眠状态和电力。

ENGLISH



冬眠舱外观 / Hibernating chamber
旧版示意，源自手册 3.0 / Historical illustration, manual 3.0

Occupied and powered

Put a Drood in the chamber and keep battery charge available. The occupant hibernates, stopping normal supply consumption and waste production. The chamber has a substantial continuous power demand. When the battery empties, normal life support consumption resumes.

Hibernation does not stop radiation

Radiation dose and radiation damage are still processed for hibernating crew. Hibernation does not grant radiation immunity. Check the compartment shield, trajectory and cumulative dose.

Plan for power loss

Keep emergency supplies for waking crew, size batteries for the longest dark period, and include RTG decay on long missions. Recheck hibernation and power after switching craft or loading a save.

16 原位资源利用与燃料合成 ISRU and fuel synthesis

中文

甲烷液氧反应器

消耗高压 CO2、水和电力，输出甲烷液氧混合燃料。准备 CO2、H2O 与 LOX/CH4 储箱，并为输出预留空间。乘员产生的是 LPCO2，必须先加压才能进入这条反应链。

ENGLISH

Methalox Generator

Consumes high-pressure CO2, water and electricity to produce methalox. Provide CO2, H2O and LOX/CH4 tanks, with room for the product. Crew exhale LPCO2, which must be compressed before entering this chain.

化学反应器模式 / Chemical Reactor mode	原料与输出 / Feedstock and product
H2O + CO2 = Methanelox	水 + 高压 CO2 + 电力，生成 LOX/CH4。 Water + high-pressure CO2 + power to LOX/CH4.
N2 + LH2 = N2H4	高压氮气 + 液氢 + 电力，生成单组元推进剂。 High-pressure nitrogen + liquid hydrogen + power to monopropellant.
LH2 + LOX = Hydrolox	液氢 + 液氧 + 电力，生成氢氧混合燃料。 Liquid hydrogen + liquid oxygen + power to hydrolox.

先选择反应，再配储箱

化学反应器提供多种模式。模式不匹配、输入耗尽、输出箱满或没有电时，都无法正常生产。游戏使用配置的转换比例；这些设备并不模拟完整的现实化工厂。

Choose the reaction before the tanks

The Chemical Reactor offers several modes. Wrong tank types, depleted feedstock, full product tanks or an empty battery prevent production. Conversion follows game settings rather than a complete real-world chemical plant.

当前采矿机隐藏于普通设计器列表，脚本主要实现展开与动画。不要把尚未完成的地表开采计入基地资源预算。

The current mining machine is hidden from the normal designer list and mainly implements deployment and animation. Do not budget for an unfinished ground-extraction system.

17 重力环与气球 Gravity rings and balloons

中文

重力环的部署

重力环位于指令类零件中。展开需要电池电量和高压氮气 N2，展开过程中消耗氮气和电力；低压氮气 LPN2 不能替代。可调整展开速度、部署旋转速度及旋转方向。

保持运行条件

完全展开并激活后，旋转持续消耗电力。保持 N2 储箱可用；当前部署判断仍检查氮气是否为空，空罐可能使设备收回。不要在展开结束时恰好耗尽氮气。

居住效果的边界

当前重力环提供可展开、旋转的居住结构。模组尚未实现独立的长期失重健康系统，因此不要把重力环计作抵消辐射、制氧或治疗的设备。乘员仍需要常规维生保障。

气球

气球位于推进类零件。激活后展开并在大气中提供与空气密度相关的向上力；不适用于真空，也不生产维生资源。先进行载荷和大气条件测试，再把它用于任务。

ENGLISH

Deploy the gravity ring

The gravity ring appears in the Command category. Deployment requires battery charge and high-pressure N2, consuming both during deployment. Low-pressure LPN2 is not a substitute. Designer controls adjust extension speed, deployment rotation speed and rotation direction.

Keep operating conditions available

Once deployed and active, rotation continuously uses electricity. Keep N2 available: the current deployment check still tests whether its source is empty, so an empty tank can cause retraction. Do not size N2 to run out exactly at the end of deployment.

Understand the habitation feature

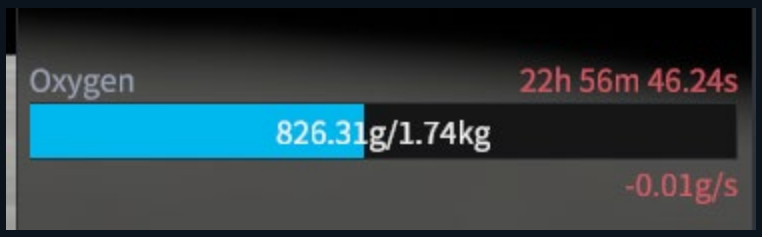
The ring provides a deployable, rotating habitat structure. This mod has no separate long-term microgravity health system in this version. The ring does not replace radiation protection, oxygen production or medical treatment. Crew still need normal life support.

Balloon

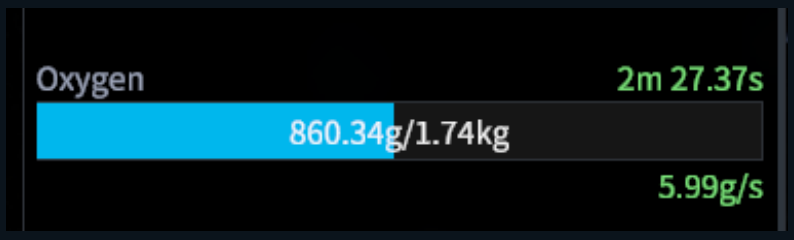
The balloon is in Propulsion. When activated, it expands and applies upward force related to air density. It provides no lift in vacuum and produces no life support resources. Test it with the intended load and atmosphere before a mission.

18 阅读飞行检查面板
Read the flight inspector

中文 ENGLISH



消耗状态：时间表示预计耗尽时间 / Consumption: time estimates depletion
旧版示意，源自手册 3.0 / Historical illustration, manual 3.0



补充状态：时间表示预计充满时间 / Refilling: time estimates time to full
旧版示意，源自手册 3.0 / Historical illustration, manual 3.0

先看变化方向

通过飞行界面的 Droodism 图标打开生命维持检查面板。飞船资源行显示当前量、总容量、净变化速率与预计时间。补给减少时看能坚持多久，补给增加时看多久充满；废物增加时应看多久储满。

Read the direction of change

Open the life support inspector from the Droodism icon in flight. Craft resource rows show amount, capacity, net rate and an estimated time. Falling supplies show endurance; rising supplies show time to full. For rising waste, read time until storage fills.

不要把当前趋势当作保证

预计时间来自最近的净变化。关闭设备、进入夜晚、切换光照模式、分离或增加乘员都会改变它。无变化时的无限或不可用显示，不代表后续永远安全。

A trend is not a guarantee

Time estimates follow the recent net change. Switching devices, entering darkness, changing lighting mode, separating or adding crew changes the estimate. An infinite or unavailable reading at zero net rate does not guarantee future safety.

同时检查每名乘员

乘员区域显示姓名、职业、任务时间、随身补给、废物及辐射。飞船总储箱有余量时，个别乘员仍可能因未正确连接共享资源而处于危险。出舱前后都应复核。

Also inspect each crew member

The crew area shows name, role, mission time, personal supplies, waste and radiation. One Drood may still be in danger if not correctly linked to shared resources even when craft tanks look healthy. Recheck before and after EVA.

19 资源预警与时间加速 Resource alerts and time warp

中文

默认开启

资源预警检查当前飞船上每名乘员的随身资源缓冲区，而不是只看全船总储量。使用外界氧气时，会跳过其氧气和 CO2 的相关预警。可在模组设置中调整阈值。

ENGLISH

Enabled by default

Alerts inspect personal buffers for each Drood on the current craft, rather than only craft-wide totals. Oxygen and CO2 checks are skipped when the Drood uses outside air. Thresholds can be changed in mod settings.

状态 State	默认触发条件 / Default trigger	处理 / Response
警告 / Warning	补给 ≤ 25%；废物 ≥ 75%。 Supply ≤ 25%; waste ≥ 75%.	显示警告；高倍加速时尝试降回实时。 Message; reduce high warp toward real time.
临界 / Critical	补给 ≤ 10%；废物 ≥ 90%。 Supply ≤ 10%; waste ≥ 90%.	显示临界提示；高倍加速时请求暂停。 Message; request pause during high warp.

收到提示后的顺序

先回到正常速度，找到提示中的乘员，补充随身资源或排出废物；再检查飞船储量、共享连接、电力和设备状态。确认恢复后再加速。

After an alert

Return to normal speed, find the named Drood, and refill supplies or offload waste. Then check craft totals, resource links, power and equipment. Resume warp only after confirming recovery.

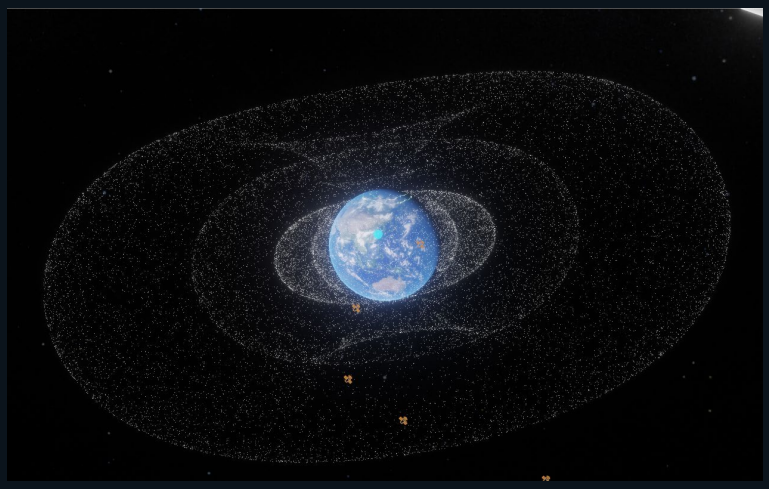
自动减速和暂停仅在代码判定的高倍加速模式中处理，且有会话防重复逻辑。提示存在冷却；它不是覆盖所有飞船与所有加速模式的无人值守保护。

Automatic slowdown and pause apply only in the high-warp modes handled by the code and use session anti-repeat flags. Messages have a cooldown. This is not unattended protection for every craft and every time mode.

20 地图中的辐射带
Radiation belts in map view

中文

ENGLISH



辐射带形状示意 / Illustration of radiation belt geometry
旧版示意，源自手册 3.0 / Historical illustration, manual 3.0

查看所在行星

在地图视图的地图信息检查面板中查看辐射带数据，并切换可见性。辐射带由行星配置生成，显示内带和外带；主菜单地图也有相应显示。

Inspect the current planet

Use the radiation belt section in the map-info inspector to read belt information and toggle visibility. Inner and outer belts are generated from planet configuration. The main-menu map also supports belt display.

路线决定暴露时间

规划穿越路线并减少在高剂量区域停留。辐射带显示用于理解空间分布；真正作用到乘员的剂量还受到舱体材料、屏蔽剩余量以及可选船内辐射源影响。

Trajectory controls exposure time

Plan crossings and reduce time spent in high-dose regions. Belt visualization helps explain spatial exposure. The dose reaching a Drood also depends on compartment material, remaining shielding and optional onboard radiation sources.

隐藏辐射带只改变显示，不会关闭辐射伤害。冬眠同样不会停止剂量累积。

Hiding the belt changes its visibility, not radiation damage. Hibernation also does not stop dose accumulation.

21 舱体辐射屏蔽 Compartment radiation shielding

中文

为乘员实际所在的舱体配置

在设计器选择舱体，设置屏蔽材料和数量。材料会影响质量、内带/外带防护及可选船内辐射防护。下表为当前游戏数值，表示剂量减免比例，不是现实材料性能。

ENGLISH

Configure the occupied compartment

Select the compartment in the designer and set shield material and amount. Material changes mass and protection against inner-belt, outer-belt and optional onboard radiation. These are current gameplay dose reductions, not real material specifications.

材料 / Material	内带 Inner	外带 Outer	船内 Onboard
无 / None	0%	0%	0%
铝 / Aluminium	20%	60%	40%
聚乙烯 / Polyethylene	80%	60%	80%
含硼聚乙烯 / Borated polyethylene	85%	65%	85%
水 / Water	75%	60%	35%
液氢 / Liquid hydrogen	95%	80%	45%
氮化硼纳米管 / Boron nitride nanotubes	80%	70%	90%

检查剩余屏蔽量

暴露会消耗屏蔽储备；它不是固定的可用小时数，也不会按舱内人数简单分摊。耗尽后不应再将其计入防护预算。当前乘员防护值在状态刷新时更新，补充或耗尽后应核对实际剂量率；出舱时不要假定仍有封闭舱防护。

Monitor the shield reserve

Exposure consumes the reserve, which is not a fixed number of hours or simply divided by occupant count. Do not budget depleted shielding as protection. Crew protection updates on state refresh, so verify actual dose rate after depletion or refilling. Do not assume compartment shielding on EVA.

水与液氢可以补充

水屏蔽和液氢屏蔽在飞行检查面板提供“补充屏蔽”。操作消耗对应飞船资源并恢复屏蔽。其他材料当前没有相同的补充按钮；补充用水会与维生和设备竞争。

Water and LH2 can be replenished

Water and liquid-hydrogen shielding offer Refill Shield in flight. It consumes the corresponding craft resource to restore protection. Other materials have no equivalent refill button in this version. Shield water competes with life support and equipment.

22 剂量、伤害与船内辐射 Dose, damage and onboard sources

中文

分清 rad 与 rad/h

rad 表示乘员已经累积的剂量；rad/h 表示当前每小时吸收速率。累计剂量随乘员保存，不会因为换船、治疗或结束一次飞行而自动清零。当前没有清除累计剂量的常规治疗。

颜色与伤害阈值不同

剂量率高于 5 rad/h 时，累计剂量达到 100、400 rad 分别触发不同伤害等级；低于 100 rad 时，剂量率高于 30 rad/h 也会受伤。达到 800 rad 后，无论当前剂量率多少都会持续受伤，急救也被禁止。表中颜色阈值不能直接当作伤害阈值。

可选船内辐射

“接收飞船内部辐射”默认关闭。开启后，当前飞船的 RTG 与已激活核热火箭发动机会增加剂量，距离和舱体屏蔽会影响结果。这是简化的游戏模型，不模拟真实核工程安全。

ENGLISH

Distinguish rad from rad/h

rad is the dose already accumulated by a crew member; rad/h is the current hourly absorption rate. Cumulative dose follows the crew record and does not automatically reset by changing craft, healing or ending a flight. No routine treatment removes accumulated dose.

界面状态 / UI state	累计剂量 / Cumulative dose	速率状态 / Dose-rate state
正常 / Nominal	< 100 rad	⩽ 1 rad/h
轻度 / Mild	100 to < 200 rad	> 1 to 5 rad/h
严重 / Severe	200 to < 500 rad	> 5 to 10 rad/h
危急 / Critical	⩾ 500 rad	> 10 rad/h

Labels and damage thresholds differ

Above 5 rad/h, cumulative doses of 100 and 400 rad trigger different damage levels. Below 100 rad, rates above 30 rad/h also cause damage. At 800 rad, damage continues regardless of current dose rate and first aid is blocked. The table's color thresholds are not the damage thresholds.

Optional onboard radiation

Receive Craft Radiation is off by default. When enabled, RTGs and activated nuclear thermal rocket engines on the current craft add dose, affected by distance and shielding. This is a simplified gameplay model rather than real nuclear engineering.

23 供电与 RTG 衰减
Power and RTG decay

中文

按最困难的时段配电

把制氧、气体压缩、污水处理、人工光照、冬眠和重力环等负载一起考虑。测试飞船最长的无光时段，并为同时启动设备预留电池。以实际设备和电池面板验证余量。

RTG 功率随任务时间下降

当前模组按每年保留 96% 的指数关系计算：P = P0 × 0.96^年数；一年采用 31,536,000 秒。它是游戏平衡参数，不等同于单纯的同位素半衰期。

ENGLISH

Size power for the hardest period

Budget electrolysis, compression, sewage treatment, artificial light, hibernation and ring loads together. Test the longest dark period and leave battery margin for simultaneous operation. Verify the reserve using the actual equipment and battery inspectors.

RTG power declines with mission age

The current model retains 96% of power per year: P = P0 × 0.96^years, using 31,536,000 seconds per year. This is a gameplay balance setting, not simply the isotope's half-life.

任务时间 / Mission age	剩余功率 / Remaining power
0 年 / years	100.0%
5 年 / years	81.5%
10 年 / years	66.5%
20 年 / years	44.2%
50 年 / years	13.0%

先预览任务末期

在设计器性能分析中选择 RTG 的预览年数，查看届时功率；飞行面板会显示任务时间、当前功率和剩余比例。若设计刚好满足出发时负载，任务后期可能失电。

Preview the end of the mission

Use Preview Years in the RTG designer performance analysis to inspect future output. Its flight inspector shows mission age, current power and remaining percentage. A design that only meets launch-day demand may lose power later.

24 模组设置速查
Mod settings reference

中文 ENGLISH

设置 / Setting	默认 Default	影响 / Effect
启用资源警告 Enable Resource Warning	开 / On	启用当前飞船乘员的资源提示与高倍加速处理。 Enable current-craft crew alerts and high-warp responses.
警告阈值 Warning Threshold	25%	可调 20%–50%；废物阈值为 1 减该值。 Range 20%–50%; waste uses one minus this value.
临界阈值 Critical Threshold	10%	可调 5%–20%；废物阈值为 1 减该值。 Range 5%–20%; waste uses one minus this value.
未加载时消耗资源 Drood Consume Resource When Unloaded	关 / Off	启用部分恢复加载时的补算；限制见第 26 章。 Enable partial catch-up on reactivation; see chapter 26.
接收飞船内部辐射 Receive Craft Radiation	关 / Off	加入当前飞船 RTG 与已激活 NTR 的辐射。 Add radiation from onboard RTGs and activated NTRs.
主动更新辐射带配置 Active Update Radiation Belt Config	关 / Off	调试时刷新运行配置，可能影响性能。 Refresh runtime configuration for debugging; may cost performance.
调试模式 Toggle Debug Mode	关 / Off	启用部分日志与隐藏的调试选项。 Enable selected logs and hidden debug controls.
备用导航球颜色 Alt Nav Ball Color	关 / Off	按速度参考系改变导航球配色。 Change navball colors by velocity reference mode.
禁用对接口相机 Disable Docking Port Camera	关 / Off	创建对接口零件时禁用其相机控制。 Disable camera control when a docking-port part is created.

设置文案可能保留旧描述；本表与资源预警章节按当前实际逻辑解释。调试开关通常应保持关闭。

Some in-game descriptions retain older wording. This table and the alert chapter describe the current logic. Debug options can normally remain off.

25 配置与自定义旗帜 Configuration and custom flags

中文

从游戏用户数据目录开始

下列路径均相对于游戏的持久化用户数据目录，不是模组工程目录。不同系统和安装方式的根路径可能不同。首次加载模组会在缺少文件时写入默认配置，通常不会覆盖已有自定义文件。

ENGLISH

Start from the game's data folder

The paths below are relative to the game's persistent user-data folder, not the mod project. The root varies by OS and installation. On load, missing default files are created, while existing custom files are normally preserved.

相对路径 / Relative path	内容 / Contents
UserData/DroodismConfig/Planets/BreathablePlanets.xml	可呼吸行星名单；使用行星的精确名称。Breathable-planet list; use exact planet names.
UserData/DroodismConfig/RadiationBeltConfigs/<planet>.xml	每个行星的辐射带形状、显示与剂量配置。Per-planet belt geometry, rendering and dose settings.
UserData/DroodismConfig/FlagImage/CustomImage.jpg	自定义旗面图片。Custom flag image.
UserData/DroodismData/CrewMembers/<save-name>.xml	按存档保存的乘员记录；应随游戏存档备份。Per-save crew records; include them in save backups.

修改名单后重启

先备份 XML，保持原有结构，仅修改所需条目。可呼吸名单有运行时缓存，修改后重启游戏再测试；名单只影响判定的一部分，稀薄空气和水下条件仍然有效。

Restart after editing the list

Back up XML files and preserve their structure. The breathable list is cached, so restart the game after editing and test again. The list only controls part of the breathing check; thin-air and underwater restrictions still apply.

替换旗面

将可读取的 JPEG 图片保存为指定的 CustomImage.jpg。已有旗帜可通过检查面板的“加载图片”操作重新读取。不要把图片路径指向项目 Assets 文件夹。

Replace the flag image

Save a readable JPEG as CustomImage.jpg at the specified location. Existing flags can reload it with Load Image in their inspector. Use the game-data location rather than the project's Assets folder.

26 读档、切船与后台消耗 Saves, switching and background use

中文

当前主要模拟正在加载的对象

正常维生、设备反应、冬眠和辐射主要依赖已加载对象的飞行更新。不要把切到另一艘船理解成整支舰队仍在完整连续模拟。

未加载消耗是有限补算

默认关闭的“未加载时消耗资源”在部分恢复物理事件中，根据经过的游戏时间补算乘员补给与废物。时间加速和进入游戏视图等原因有专门排除；它不覆盖所有加载、切换与读档路径。

设备不会完整回放后台过程

补算不能保证同时重现发电、日夜循环、反应器、冬眠、屏蔽损耗或完整的后台健康结果。长任务仍应携带余量，返回飞船后以面板重新确认资源、废物、电力和辐射。

备份乘员记录与游戏存档

累计辐射等乘员数据另存于按存档命名的 XML，并与游戏乘员记录同步。迁移、恢复或分享任务时，连同对应 CrewMembers 文件备份；避免在正在运行的游戏中手动改写乘员数据。

独立的全功能后台模拟器尚未接入。手册中的计划任务建议应结合这些限制使用。

ENGLISH

Simulation centers on loaded objects

Normal life support, equipment reactions, hibernation and radiation mainly depend on flight updates for loaded objects. Switching craft does not imply a complete continuously simulated fleet.

Unloaded consumption is partial catch-up

The off-by-default unloaded-consumption setting applies elapsed-time supply and waste catch-up on some physics reactivation events. Warp and loaded-into-view reasons have explicit exclusions. It does not cover every loading, switching or save path.

Equipment history is not fully replayed

Catch-up does not guarantee replay of power generation, day/night cycles, reactors, hibernation, shield depletion or complete background health outcomes. Carry margin and recheck supplies, waste, power and radiation when returning to a craft.

Back up crew records with saves

Crew information such as cumulative radiation is stored in a per-save XML file and synchronized with the game's crew records. Include the corresponding CrewMembers file when migrating or restoring missions. Avoid rewriting crew data while the game is running.

A separate full-featured background simulator is not connected in this version. Apply mission-planning advice with these limits in mind.

27 常见问题排查 Troubleshooting

中文 ENGLISH

现象 / Symptom	优先检查 / First checks
模组启用失败 Mod fails to initialize	Juno Harmony 是否已安装并先启用；是否有旧模组冲突。 Check Juno Harmony installation and activation, then old or conflicting mods.
液氧很多但乘员缺氧 Plenty of LOX, but no oxygen	LOX 与 Oxygen 不同；增加对应减压装置及气态氧气容量。 LOX is not Oxygen. Add the correct decompressor and low-pressure storage.
登船后仍然告警 Alert persists after boarding	检查该乘员随身缓冲区、所在飞船补给、共享引用和废物空间。 Check personal buffers, the occupied craft's supplies, resource links and waste room.
处理设备不生产 A processor produces nothing	是否激活；电量、原料、资源类型、输出空位和展开状态。 Check activation, charge, feedstock, tank types, output room and deployment.
生物反应器不产食物 No food from the bioreactor	进度是否完成；光照指向、人工光、电池、CO2、水和废水容量。 Check batch progress, lighting, battery, CO2, water and wastewater room.
无法治疗 Healing is refused	选中 Drood 且距离 ⩽ 5 m；有治疗点；无空补给、满废物或 ⩾ 800 rad。 Select a Drood within 5 m; check kit points, personal supplies, waste and cumulative dose.
降落伞无法打开 Parachute will not deploy	舱外、离地 > 10 m、未入水、空气足够、低于音速；自动模式还需下降。 Check EVA, height > 10 m, water, air density and subsonic speed; auto also needs descent.
冬眠中仍受伤 Damage during hibernation	检查辐射；电池耗尽也会恢复正常消耗。 Check radiation; an empty battery also restores normal consumption.

28 任务前后的检查清单 Mission checklists

中文

短途与首次试飞

1. 核对乘员人数和职业。
2. 按乘员天数准备资源包或独立储箱。
3. 保留三种废物的空位。
4. 在地面观察净消耗与单人资源。
5. 做一次登船、出舱、保存与读取测试。

长期飞行与空间站

1. 配置氧气生产或液氧减压。
2. 决定 CO2 的排放或利用方式。
3. 保留清水与食物库存。
4. 测试最长无光期及 RTG 末期功率。
5. 检查屏蔽材料和累计辐射。
6. 准备急救、工具与返航余量。

基地、切船与恢复任务

1. 只把可验证的水源计入预算。
2. 为反应器配置正确的高低压资源。
3. 检查分离后乘员所在一侧的资源。
4. 依据后台模拟限制安排无人值守时间。
5. 返回飞船后先看面板，再加速。

储量、生产速率、电力和废物空间需要同时满足。任何一项先到极限，都会决定任务能否继续。

ENGLISH

Short missions and first tests

1. Confirm crew count and roles.
2. Size packs or tanks by crew-days.
3. Leave room for all three wastes.
4. Observe net rates and personal resources on the ground.
5. Test boarding, EVA, saving and loading.

Long flights and stations

1. Provide oxygen production or LOX decompression.
2. Decide whether to vent or reuse CO2.
3. Carry water and food reserves.
4. Test the longest darkness and end-of-mission RTG power.
5. Check shielding and cumulative dose.
6. Bring medical supplies, tools and return margin.

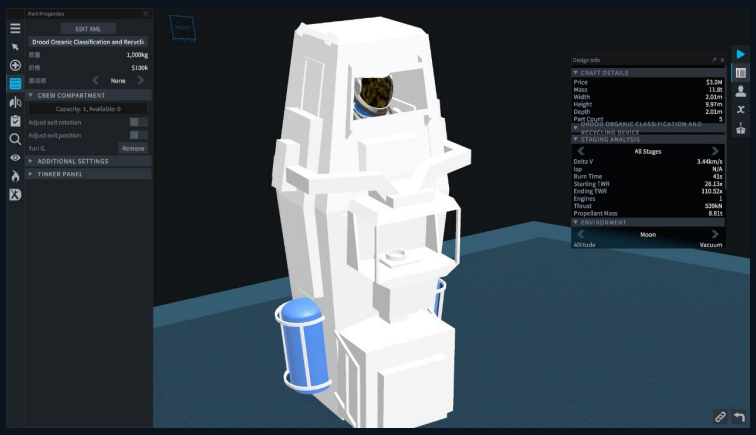
Bases, switching and resuming

1. Budget only verified water sources.
2. Match high- and low-pressure feedstocks to reactors.
3. Check the crew's side after separation.
4. Plan unattended periods around background limits.
5. Inspect the craft on return before accelerating time.

Supply reserves, production rate, power and waste capacity must all be sufficient. The first limit reached can decide whether the mission continues.

29 附录：有机分类与回收装置 Appendix: the DOCR device

中文 ENGLISH



Drood 有机分类与回收装置 / Drood Organic Classification and Recycling Device
旧版示意，源自手册 3.0 / Historical illustration, manual 3.0

一个黑色幽默设备

DOCR 装置保留了旧手册末尾的黑色幽默设定。它在装入一名乘员并被激活后，持续伤害该乘员，同时回收其随身资源并向飞船提供部分资源。

A dark-humor device

The DOCR device preserves the dark-humor feature from the end of the old manual. With one occupant and the device activated, it damages the Drood while recovering carried supplies and adding some resources to the craft.

使用前理解后果

这不是冬眠舱或急救设备，可能导致乘员死亡。若要恢复受伤乘员，应先解除维生问题并使用急救包；不能沿用旧手册中“飞行中无法治疗”的说法来判断是否应回收乘员。

Understand the consequence

This is not a hibernation chamber or medical device and can kill the occupant. To recover an injured Drood, first resolve life support problems and use first aid. The old claim that crew cannot be healed in flight is no longer valid.

> 这是游戏中的可选内容，不是标准维生设计的一部分。正式任务中应避免误激活。
>
> This is an optional in-game feature rather than part of a standard life support design. Avoid accidental activation during a mission.

30 零件与章节索引 Parts and chapter index

中文 ENGLISH

设计器分类 Category	零件 / Parts	章节 Ch.
维生资源 Life Support	资源包、资源箱、食物箱、水箱 Resource pack, resource fuselage, food tank, water tank	02, 03, 08, 11
维生资源 Life Support	电解水装置、CO2 过滤器 Electrolytic device, CO2 filter	09
维生资源 Life Support	污水处理装置、过氧化钠容器 Sewage treatment, sodium peroxide container	10, 11
维生资源 Life Support	气体减压装置、气体加压装置 Gas depressurizer, gas pressurizer	13
维生资源 Life Support	急救包、工具箱、DOCR 装置 First aid kit, tool kit, DOCR device	06, 29
原位资源利用 ISRU	光生物反应器、水泵 Photobioreactor, water pump	10, 12
原位资源利用 ISRU	甲烷液氧反应器、化学反应器 Methalox generator, chemical reactor	16
指令 Command	冬眠舱、重力环 Hibernating chamber, gravity ring	15, 17
推进 Propulsion	气球 / Balloon	17
原版零件扩展 Stock part extensions	乘员、燃料电池、RTG、载人舱 Droods, fuel cells, RTGs, crew compartments	04–07, 14, 21–23

个人伞和旗帜由相应玩法创建，不作为普通可选零件列出。采矿机在当前普通设计器中隐藏。

Personal canopies and flags are created by their gameplay actions rather than offered as ordinary selectable parts. The mining machine is hidden in the current normal designer.

31 版本说明与致谢
Edition notes and credits

中文

本版范围

手册 4.0，适用本地项目中标记为 0.89 的模组，日期 2026-10-24。功能、数值、操作条件与限制以当前工作区代码和零件定义为准；本地化用于核对名称。旧手册仅提供图片与原版致谢。本次未进行游戏内全流程验收。

相对 3.0 的主要修订

旧版适用模组 0.795。本版改为中英对照，重组重复和空白页，修正治疗与呼吸条件，补充资源预警、默认设置、RTG 衰减、重力环、配置路径、后台限制及故障排查；不把计划中的降落伞移植和后台系统写成已完成功能。

截图说明

封面与历史截图来自用户提供的 3.0 手册，只用于识别外观和理解界面，均标明为旧版示意。图片中的版本号、数值、标签和布局可能与当前游戏不同，以本版文字及游戏实测为准。

保留原版致谢

模组作者：SatelliteTorifune。
模型：w2semail、TN920、DeepSpaceIndustries。

模组缩略图：TN920。
测试：Charodey、GreenGalaxyAeroSpace。
感谢每位玩家的反馈与飞行记录。

ENGLISH

Scope of this edition

Manual 4.0 targets the local project marked mod 0.89, dated 2026-10-24. Current working-tree code and part definitions determine features, values, conditions and limits; localization supplies labels. The old manual supplies images and original credits only. No end-to-end in-game acceptance test was performed.

Main revisions from edition 3.0

Edition 3.0 targeted mod 0.795. This edition adds parallel Chinese and English, removes repeated and blank pages, corrects healing and breathing guidance, and covers alerts, settings, RTG decay, the gravity ring, configuration, background limits and troubleshooting. Planned parachute and background-system ports are not presented as completed features.

Screenshot provenance

The cover and historical screenshots come from the supplied edition 3.0 manual and illustrate appearance and UI concepts. Screenshot version numbers, values, labels and layouts may differ from the current game; follow this edition's text and actual in-game readings.

Credits retained from the original

Mod author: SatelliteTorifune.
Models: w2semail, TN920, DeepSpaceIndustries.
Mod thumbnail: TN920.
Testing: Charodey, GreenGalaxyAeroSpace.
Thanks to every player contributing feedback and flight experience.